Swapnamoy Garai

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Professional Summary

Agentic AI and Backend Systems Engineer pursuing B.Tech in CSE at Jadavpur University, currently interning at Quantiphi Analytics. Proven expertise in developing **Model Context Protocol (MCP)** tool execution engines, **autonomous coding agents** with reflection/self-correction loops, and **AST-driven execution sandboxes**. Proficient in high-concurrency microservices in **Python** and **Go**, distributed locking, and resilient API architectures.

Technical Skills

Agentic AI & LLMs: **Model Context Protocol (MCP)**, Autonomous Agent Loops, **AST Security Scanners**, Self-Correction & Reflection, Tool Calling, **Semantic Caching**, Claude API, OpenAI API
Languages: **Python**, **Go (Golang)**, **C++**, **JavaScript/TypeScript**, SQL, Rust, C
Backend & Cloud Systems: **FastAPI**, **Docker**, **gRPC**, **Protobuf**, **Redis**, PostgreSQL, Token Bucket Rate Limiting, Circuit Breakers, Distributed Locking
Developer Tools & Practices: Git, GitHub CLI, Linux/WSL, CI/CD Pipelines, Test-Driven Development (**TDD**), Pytest, k6

Work Experience

Quantiphi Analytics Bengaluru, India
Software Developer Intern *Jun 2026 – Present*

- Engineered production **FastAPI** REST microservices using contract-first **OpenAPI** specifications backed by **PostgreSQL** and **Redis**.
- Applied **Test-Driven Development (TDD)** authoring comprehensive unit and integration suites to sustain an **85%+** CI/CD pass rate.
- Designed dimensional star schemas and optimized transformation SQL with window functions for enterprise data pipelines.
- Configured automated PR gating checks and containerized workflows to ensure code quality and deployment stability.

Crispat Remote
Full Stack Developer Intern *Jun 2024 – Sep 2024*

- Developed full-stack features with **React.js**, **Node.js**, and **Express**, utilizing **Redis** for real-time state tracking.
- Built a scalable asynchronous multipart file pipeline, reducing research document upload latency by **50%**.
- Created interactive connection graph visualizations with **React Flow** and **D3.js**, enhancing user engagement by **40%**.

Featured Projects

Py-Sandbox-Agent | GitHub | *Autonomous Coding Agent & Isolated Sandbox* Python, Docker, AST, FastAPI

- Engineered an autonomous coding agent with a **reflection/self-correction FSM loop** that parses test tracebacks and iteratively refines generated code up to `max_retries`.
- Implemented an **AST static security scanner** enforcing default-deny import filtering (`os`, `subprocess`), blocking unsafe builtins (`eval`, `exec`), and preventing dunder-attribute escapes prior to execution.
- Enforced multi-tenant container isolation via network-gapped (`network=none`), read-only root, and **256MB memory caps** in Docker, governed by a **100-slot concurrency pool** with HTTP 503 backpressure.

Agentexec Engine | GitHub | *Asynchronous MCP Tool-Execution Engine* Go (Golang), gRPC, Protobuf, Redis

- Architected a concurrent Go worker engine executing **Model Context Protocol (MCP)** tool calls with **distributed locking**, exponential backoff retry (50ms–1s), and **atomic idempotency reservation**.
- Implemented a per-tool 3-state **Circuit Breaker** (Closed/Open/Half-Open) and an in-memory **semantic cache** with TTL and canonicalized key hashing to eliminate redundant tool invocations.
- Enforced hardware resource bounds via POSIX `ulimit` memory ceilings and process-group `SIGKILL` management to protect host workers from runaway executions.

Codeforces Problem Recommender | GitHub | *Browser Extension & Scalable State Engine* JavaScript, Chrome/Web APIs

- Engineered a competitive programming browser extension using clean **OOP design patterns** and decoupled background service workers.
- Built a persistent multi-problem timer system using **browser.alarms**, achieving **80% higher accuracy** than standard Date-object polling.
- Implemented smart client-side **caching** for Codeforces API responses, yielding sub-millisecond (near 0ms) subsequent load times.

Education

Jadavpur University Kolkata, India
B.Tech in Computer Science & Engineering ***(CGPA: 8.4 / 10.0)*** *Nov 2022 – June 2026*

Honors & Certifications

Competitive Programming: Solved **800+ LeetCode problems** (Peak Rating: **2000**, Best Contest Rank: **208**). Active on Codeforces & AtCoder.
Standardized Exams: **JEE Advanced AIR 5576** | **WBJEE Rank 207** (Top 0.2% among 100,000+ candidates).
Certification: **Claude Certified Developer** – Foundations Certified.